## EXPERIMENT NO: 15

Experiment to understand the data preprocessing in Data science

**PROGRAM:**

```
import numpy as np
import pandas as pd
df=pd.read_csv("pre-process_datasample.csv")
print(df)
print(df.info())
print(df.Country.mode())
print(df.Country.mode()[0])
print(type(df.Country.mode()))
df.Country.fillna(df.Country.mode()[0],inplace=True)
df.Age.fillna(df.Age.median(),inplace=True)
df.Salary.fillna(round(df.Salary.mean()),inplace=True)
print(df)
print(pd.get_dummies(df.Country))
updated_dataset=pd.concat([pd.get_dummies(df.Country),df.iloc[:,[1,2,3]]],axis=1)
print(updated_dataset)
print(df.info())
updated_dataset.Purchased.replace(['No','Yes'],[0,1],inplace=True)
print(updated_dataset)
```

OUTPUT:

| | Country | Age | Salary | Purchased |
|---|---|---|---|---|
| **0** | France | 44.0 | 72000.0 | No |
| **1** | Spain | 27.0 | 48000.0 | Yes |
| **2** | Germany | 30.0 | 54000.0 | No |
| **3** | Spain | 38.0 | 61000.0 | No |
| **4** | Germany | 40.0 | NaN | Yes |
| **5** | France | 35.0 | 58000.0 | Yes |
| **6** | Spain | NaN | 52000.0 | No |
| 7 | France | 48.0 | 79000.0 | Yes |
| **8** | Germany | 50.0 | 83000.0 | No |
| **9** | France | 37.0 | 67000.0 | Yes |

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    10 non-null     object
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes
```

| | Country |
|---|---|
| **0** | France |

**dtype:** object

'France'

**pandas.core.series.Series**

def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None, fastpath: bool | lib.NoDefault=lib.no_default) -> None

One-dimensional ndarray with axis labels (including time series).

Labels need not be unique but must be a hashable type. The object supports both integer- and label-based indexing and provides a host of methods for performing operations involving the index. Statistical methods from ndarray have been overridden to automatically exclude

| | Country | Age | Salary | Purchased |
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| **1** | Spain | 27.0 | 48000.0 | Yes |
| **2** | Germany | 30.0 | 54000.0 | No |
| **3** | Spain | 38.0 | 61000.0 | No |
| **4** | Germany | 40.0 | 63778.0 | Yes |
| **5** | France | 35.0 | 58000.0 | Yes |
| **6** | Spain | 38.0 | 52000.0 | No |
| 7 | France | 48.0 | 79000.0 | Yes |
| **8** | Germany | 50.0 | 83000.0 | No |
| **9** | France | 37.0 | 67000.0 | Yes |

| | France | Germany | Spain |
|---|---|---|---|
| **0** | True | False | False |
| **1** | False | False | True |
| **2** | False | True | False |
| **3** | False | False | True |
| **4** | False | True | False |
| **5** | True | False | False |
| **6** | False | False | True |
| 7 | True | False | False |
| **8** | False | True | False |
| **9** | True | False | False |

| | France | Germany | Spain | Age | Salary | Purchased |
|---|---|---|---|---|---|---|
| **0** | True | False | False | 44.0 | 72000.0 | No |
| **1** | False | False | True | 27.0 | 48000.0 | Yes |
| **2** | False | True | False | 30.0 | 54000.0 | No |
| **3** | False | False | True | 38.0 | 61000.0 | No |
| **4** | False | True | False | 40.0 | 63778.0 | Yes |
| **5** | True | False | False | 35.0 | 58000.0 | Yes |
| **6** | False | False | True | 38.0 | 52000.0 | No |
| 7 | True | False | False | 48.0 | 79000.0 | Yes |
| **8** | False | True | False | 50.0 | 83000.0 | No |
| **9** | True | False | False | 37.0 | 67000.0 | Yes |

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    10 non-null     object
 1   Age        10 non-null     float64
 2   Salary     10 non-null     float64
 3   Purchased  10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes
```

| | France | Germany | Spain | Age | Salary | Purchased |
|---|---|---|---|---|---|---|
| **0** | True | False | False | 44.0 | 72000.0 | 0 |
| **1** | False | False | True | 27.0 | 48000.0 | 1 |
| **2** | False | True | False | 30.0 | 54000.0 | 0 |
| **3** | False | False | True | 38.0 | 61000.0 | 0 |
| **4** | False | True | False | 40.0 | 63778.0 | 1 |
| **5** | True | False | False | 35.0 | 58000.0 | 1 |
| **6** | False | False | True | 38.0 | 52000.0 | 0 |
| 7 | True | False | False | 48.0 | 79000.0 | 1 |
| **8** | False | True | False | 50.0 | 83000.0 | 0 |
| **9** | True | False | False | 37.0 | 67000.0 | 1 |